ECONOMETRICS

BRUCE E. HANSEN

©2000, 2014[1]

University of Wisconsin
Department of Economics

This Revision: January 3, 2014
Comments Welcome

[1]This manuscript may be printed and reproduced for individual or instructional use, but may not be printed for commercial purposes.

Contents

Preface . . . viii

1 Introduction 1

1.1 What is Econometrics? . . . 1

1.2 The Probability Approach to Econometrics . . . 1

1.3 Econometric Terms and Notation . . . 2

1.4 Observational Data . . . 3

1.5 Standard Data Structures . . . 4

1.6 Sources for Economic Dat . . . 5

1.7 Econometric Softwa . . . 6

1.8 Reading the Manusc . . . 7

1.9 Common Symbols . . . 8

2 Conditional Expectation and Projection 9

2.1 Introduction . . . 9

2.2 The Distribution of Wages . . . 9

2.3 Conditional Expectation . . . 11

2.4 Log Differences* . . . 13

2.5 Conditional Expectation Function . . . 14

2.6 Continuous Variables . . . 15

2.7 Law of Iterated Expectations . . . 16

2.8 CEF Error . . . 18

2.9 Intercept-Only Model . . . 19

2.10 Regression Variance . . . 20

2.11 Best Predictor . . . 20

2.12 Conditional Variance . . . 21

2.13 Homoskedasticity and Heteroskedasticity. . . . 23

2.14 Regression Derivative . . . 23

2.15 Linear CEF . . . 24

2.16 Linear CEF with Nonlinear Effects . . . 25

2.17 Linear CEF with Dummy Variables . . . 26

2.18 Best Linear Predictor . . . 28

2.19 Linear Predictor Error Variance . . . 34

2.20 Regression Coefficients . . . 35

2.21 Regression Sub-Vectors . . . 35

2.22 Coefficient Decomposition . . . 36

2.23 Omitted Variable Bias . . . 37

2.24 Best Linear Approximation . . . 38

2.24 Best Linear Approximation . . . 38

2.25 Normal Regression . . . 39

2.26 Reverse Regression . . . 40

2.27 Limitations of the Best Linear Predictor . . . 41

2.29 Random Coefficient Model 2.30 41
2.30 Causal Effects 43
2.31 Expectation: Mathematical Details* 47
2.32 Existence and Uniqueness of the Conditional Expectation* 49
2.33 Identification* 50
2.34 Technical Proofs* 51
Exercises 55

3 The Algebra of Least Squares 57
3.1 Introduction 57
3.2 Random Samples 57
3.3 3.4 Sample Means 58
Least Squares Estimator 58
3.5 Solving for Least Squares with One Regressor 59
3.6 Solving for Least Squares with Multiple Regressors 60
3.7 Illustration 62
3.8 Least Squares Residuals 62
3.8 3.9 Squares Residuals . . . . . .
3.9 Matrix Notation 65
3.10 Projection Matrix 65
3.11 Orthogonal Projection 66
3.12 Estimation of Error Variance 67
3.13 Analysis of Variance 68
3.14 Regression Components 68
3.15 Residual Regression 70
3.16 Prediction Errors 71
3.17 Influential Observations 72
3.18 Normal Regression Model 74
3.19 CPS Data Set 76
3.20 Programming 78
3.21 Technical Proofs* 82
Exercises 83
4 Least Squares Regression
4.1 Introduction 86
4.1 Introduction 86
4.2 Sample Mean 86
4.3 87
4.4 Mean of Least-Squares Estimator 88
4.5 89
4.6 Variance of Least Squares Estimator 91
4.7 Gauss-Markov Theorem . . . . . .
Residuals 92
4.8 Estimation of Error Variance 93
4.9 Mean-Square Forecast Error 95
4.10 Covariance Matrix Estimation Under Homoskedasticity 96
4.11 Covariance Matrix Estimation Under Heteroskedasticity 97
4.12 Standard Errors 100
4.13 Computation 101
4.14 Measures of Fit 102
4.15 Empirical Example 103
4.16 Multicollinearity 105
4.17 Normal Regression Model 108
Exercises 110